

$$\equiv \forall x \forall y (A(x,y) \rightarrow \neg A(x,x)) \quad (\text{Ex-1.3 Prob 21})$$

$$d) \forall y \exists x \exists z (T(x,y,z) \vee A(x,y))$$

$$\neg \forall y \exists x \exists z (T(x,y,z) \vee A(x,y)) \equiv \exists y \neg \exists x \exists z (T(x,y,z) \vee A(x,y))$$

$$(\text{De Morgan's Law of quantifiers}) \equiv \exists y \forall x \neg \exists z (T(x,y,z) \vee A(x,y))$$

$$(\text{De Morgan's Law of quantifiers}) \equiv \exists y \forall x \forall z \neg (T(x,y,z) \vee A(x,y))$$

$$(\text{De Morgan's Law of quantifiers}) \equiv \exists y \forall x \forall z (\neg T(x,y,z) \wedge \neg A(x,y))$$

$$(\text{2nd De Morgan's Law})$$

33) Rewrite each of these statements so that negations appear only within predicates (that is, so that no negation is outside a quantifier or an expression involving logical connectives).

$$a) \neg \forall x \forall y P(x,y) \equiv \exists x \neg \forall y P(x,y) \quad (\text{De Morgan's Law of quantifiers})$$

$$\equiv \exists x \exists y \neg P(x,y) \quad (\text{De Morgan's Law of quantifiers})$$

$$b) \neg \forall y \exists x P(x,y) \equiv \exists y \neg \exists x P(x,y) \quad (\text{De Morgan's Law of quantifiers})$$

$$\equiv \exists y \forall x \neg P(x,y) \quad (\text{De Morgan's Law of quantifiers})$$

$$c) \neg \forall y \forall x (P(x,y) \vee A(x,y)) \equiv \exists y \neg \forall x (P(x,y) \vee A(x,y)) \quad (\text{De Morgan's Law of quantifiers})$$

$$\equiv \exists y \exists x \neg (P(x,y) \vee A(x,y)) \quad (\text{De Morgan's Law of quantifiers})$$

$$\equiv \exists y \exists x (\neg P(x,y) \wedge \neg A(x,y)) \quad (\text{2nd De Morgan's Law})$$

$$d) \neg (\exists x \exists y \neg P(x,y) \wedge \forall x \forall y A(x,y)) \equiv \neg \exists x \exists y \neg P(x,y) \vee \neg \forall x \forall y A(x,y)$$

$$(\text{2nd De Morgan's Law}) \equiv \forall x \neg \exists y \neg P(x,y) \vee \exists x \neg \forall y A(x,y) \quad (\text{2 applications of De Morgan's Law of quantifiers})$$

$$\equiv \forall x \forall y \neg (\neg P(x,y)) \vee \exists x \exists y \neg A(x,y) \quad (\text{Double negation law})$$

$$e) \neg \forall x (\exists y \forall z P(x,y,z) \wedge \exists z \forall y P(x,y,z)) \equiv \exists x \neg (\exists y \forall z P(x,y,z) \wedge \exists z \forall y P(x,y,z))$$

$$\equiv \exists x (\neg \exists y \forall z P(x,y,z) \vee \neg \exists z \forall y P(x,y,z)) \quad (\text{De Morgan's Law of quantifiers})$$

$$\equiv \exists x (\forall y \neg \forall z P(x,y,z) \vee \forall z \neg \forall y P(x,y,z)) \quad (\text{1st De Morgan's Law})$$

$$\equiv \exists x (\forall y \exists z \neg P(x,y,z) \vee \forall z \exists y \neg P(x,y,z)) \quad (\text{2 applications of De Morgan's Law of quantifiers})$$